

Question 1:

The total parameters of a model are:

$$(3DF + 4DNH + D) \cdot L + D \cdot V$$

If we assume $F = 4 \cdot D$, $NH = D$ we get:

$$(16D^2 + D) \cdot L + D \cdot V$$

Let $D = 4096$, $V = 32000$, $L = 64$. Hence

we get $(16 \cdot (4096)^2 + 4096) \cdot 64 +$

$$4096 \cdot 32000 \simeq 17.3 \cdot 10^9$$

$\simeq 17.3$ Billion parameters.

Attention params are $4 \cdot D^2 \cdot L$ hence

$$4 \cdot (4096)^2 \cdot 64 \simeq 4.3 \text{ Billion, so the}$$

fraction is $\frac{4.3}{17.3} \simeq 0.24$.

The size of the KV cache is $2 \cdot S \cdot L \cdot N \cdot H$
 and since $N \cdot H = D$, we get $2 \cdot L \cdot D$ per token
 i.e. $2 \cdot (64) \cdot 4096 \approx 524 \text{ KB}$.

Question 2:

$$\text{let Mesh} = \{x:4, y:8, z:4\}.$$

Assume the intended operation is

$$A \in \mathbb{R}^{B \times D_y} \cdot W \in \mathbb{R}^{D_y \times F}. \text{ Each}$$

device (i, j, k) performs $2 \cdot \frac{B}{|x|} \cdot \frac{D}{|y|} \cdot F$ flops

hence total flops is :

$$\begin{aligned} \text{Total FLOPs} &= \cancel{|x|} \cdot \cancel{|y|} \cdot |z| \cdot 2 \cdot \frac{B \cdot D \cdot F}{\cancel{|x|} \cdot \cancel{|y|}} \\ &= 8 B D F. \end{aligned}$$

Question 3:

Assume the intended operation is

$$A[I, J, K, L] \cdot B[I, J, M, N, O] \rightarrow C[K, L, M, N, O]$$

by means of:

$$C[K, L, M, N, O] = \sum_{j=1}^J \sum_{i=1}^I A[i, j, K, L] \cdot B[i, j, M, N, O]$$

we do $K \cdot L \cdot M \cdot N \cdot O \cdot 2 \cdot I \cdot J$.

Question 4:

Assume our naive attention algorithm is writes every intermediate result back to HBM and also that $K=N$. Hence we do:

$$1. L[B, T, S, N] \leftarrow Q[B, T, N, H] \cdot K[B, S, N, H]$$

$$2. S[B, T, S, N] \leftarrow \text{SoftMax}_S(L[B, T, S, N])$$

$$3. [B, T, N, H] \leftarrow S[B, T, S, N] \cdot V[B, S, N, H]$$

Assume that $N \cdot H = D$.

$$\text{Bytes step 1} = 2BDTNH + 2BDSNH + 2BTSN$$

$$= 2BTD + 2BSD + 2BTSN$$

$$\text{FLOPS step 1} = 2BNTSH = 2BTS D$$

$$\text{Bytes step 2} = 2BTSN + 2BTSN = 4BTSN$$

$$\text{FLOPS step 2} = O(BTSN)$$

$$\begin{aligned}\text{Bytes Step 3} &= 2 \cdot B T S N + 2 B S N H + 2 B T N H \\ &= 2 B T S N + 2 B S D + 2 B T D\end{aligned}$$

$$\begin{aligned}\text{FLOPs Step 3} &= 2 B N T H S \\ &= 2 B T S D\end{aligned}$$

Assume Flash Attention so $L[B, T, S, N]$ does not need to be in HBM;

$$\text{Bytes Total} = 4 \cdot (B S D + B T D)$$

$$\text{FLOPs Total} = 2 B T S D + 2 B T S D = 4 B T S D$$

$$AI(T, S) = \frac{B T S D}{B S D + B T D} = \frac{T S D}{S D + T D}$$

$$= \frac{T S}{T + S} = \frac{T S}{T + S}$$

Let $S = T$:

$$A_I = \frac{T^2}{2 \cdot T} = \frac{T}{2}$$

If $T \gg 2 \cdot A_I$ (Accelerator)

we are FLOPs bound. If we assume
VSe then $A_I(VSe) \approx 240$ hence
 $T \gg 480$.

It by effective relative cost they mean

$$\text{Effective cost}(T) = \frac{t_{\text{Attention}}(T)}{t_{\text{FFW}}(T)}$$

where T is context length not time.

Assume $B > 240$ so that in the FFW
block we become compute bound, hence

$$t_{\text{FFW}} = \frac{6 B T D F}{\text{FLOPs/s}}$$

$$t_{\text{attention}} = \max\left(\frac{8BD}{BW}, \frac{4BT^2D}{\text{FLOPs/s}}\right)$$

$$\Rightarrow \text{Effective cost}(T) = \max\left(\frac{8BD}{BW}, \frac{4BT^2D}{\text{FLOPs/s}}\right)$$

$$\frac{6BDTF}{\text{FLOPs/s}}$$

$$= \max\left(\frac{8BD}{BW}, \frac{4BT^2D}{\text{FLOPs/s}}\right) \cdot \frac{\text{FLOPs/s}}{6BDTF}$$

If $T \geq 480$ then:

$$\text{Effective cost}(T) = \frac{4BT^2D}{6BDTF} = \frac{4T}{6F}$$

$$= \frac{4 \cdot T}{24D} = \frac{T}{6D}$$

If $T < 480$ then:

$$\text{Effective cost (T)} = \frac{8 \text{ BTD}}{\text{BW}} \cdot \frac{\text{FLOPs/s}}{6 \text{ BTD F}}$$

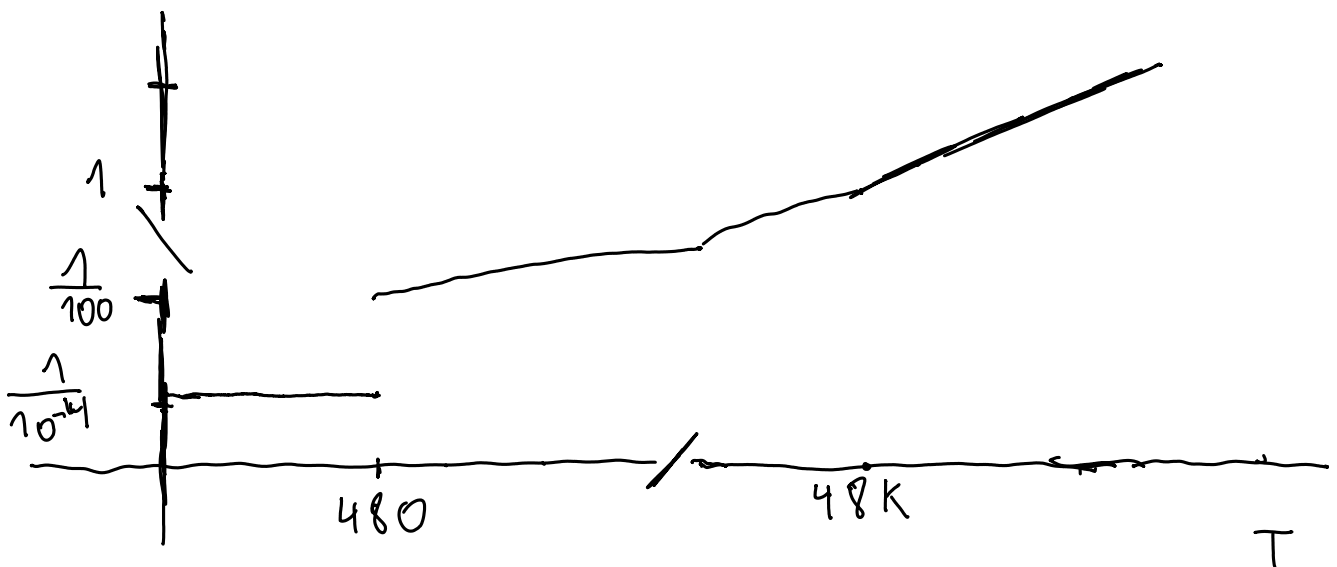
$$= \frac{8 \cancel{\text{BTD}}}{6 \cancel{\text{BTD}} \cancel{\text{F}}} \cdot \frac{1}{240} = \frac{8}{6 \text{ F}} \cdot \frac{1}{240}$$

$$= \frac{8}{240} \cdot \frac{1}{240} = \frac{8}{24 \cdot 240 \cdot 0} = \frac{8}{5760 \cdot 0}$$

$$\approx \frac{13}{10^{-4}} \cdot \frac{1}{0}$$

Assume $0 \approx 8 \text{ k}$.

Effective cost (T)



Question 5:

Assuming flops without training, $D = N \cdot H$, $S = T$ and $N = K$, then we have to solve for:

$$4 B T^2 D = 8 B T D N H$$

$$\Rightarrow 4 \cancel{B} T^2 D = 8 \cancel{B} T D^2$$

$$\Rightarrow T = \frac{8}{4} D \Rightarrow T = 2 D.$$

So a $T = 2D$ sequence length self-attention FLOPs equal the QKVO projection FLOPs.

Question 6:

Define a general layer as:

$$O = f(I, W)$$

In a computational graph, during the backward pass, it will receive

$\frac{dL}{dO}$, and it will compute

$\frac{dL}{dI}$, $\frac{dL}{dW}$ and send $\frac{dL}{dI}$ down

the graph. Note that in general

$$\frac{dL}{dW} = \frac{dL}{dO} \cdot \frac{dO}{dW}(W, I) \text{ so it depends}$$

on I as well. With this in mind we can see that if we draw the computational graph we will need to rematerialize:

$$I = \text{gelu}(x \cdot W_{in1}) \odot (x \cdot W_{in2}) \text{ for } \frac{dL}{dW_{out}}$$

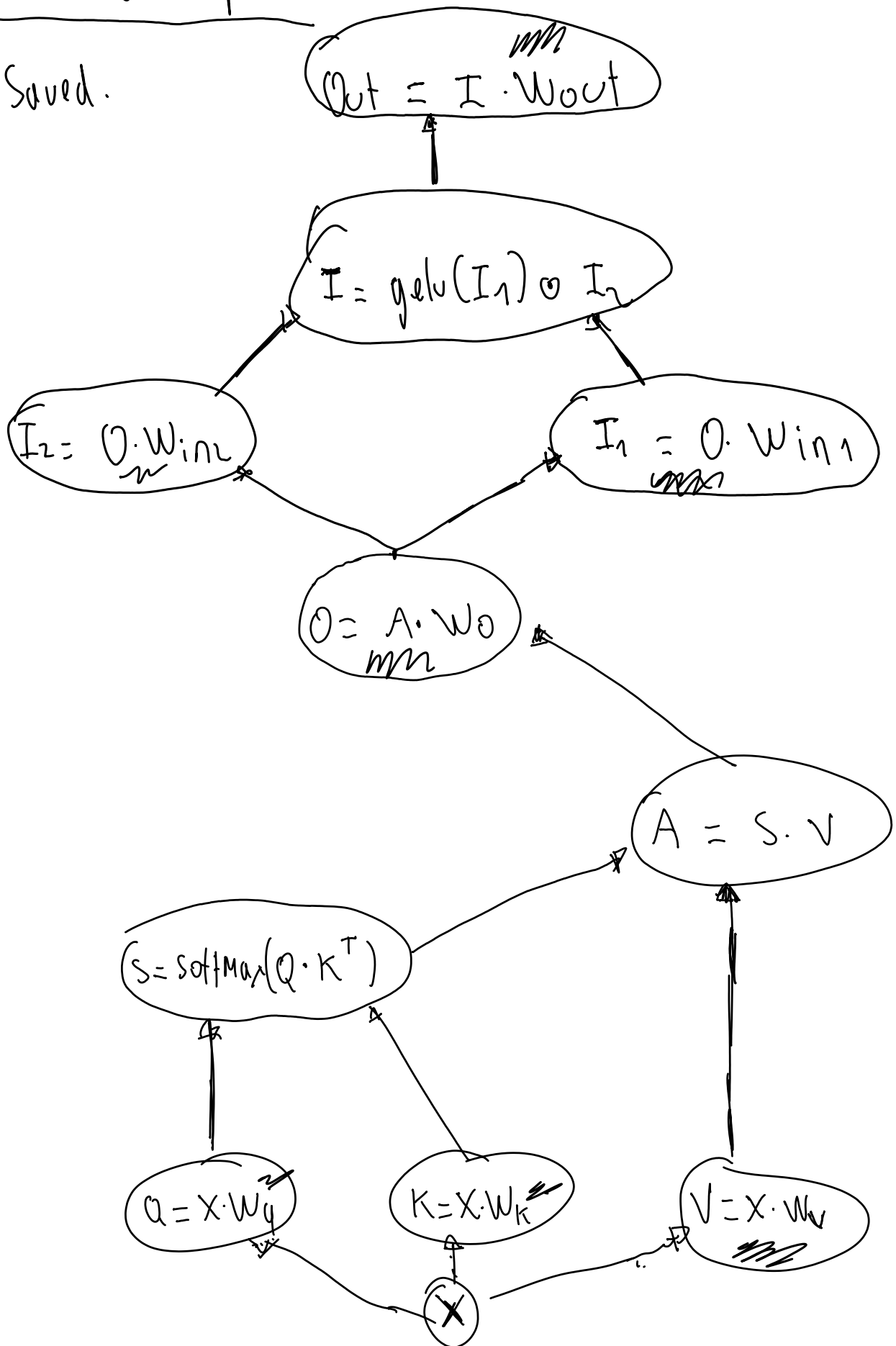
$$A = \text{softmax}(Q \cdot K^T) \cdot V \text{ for } \frac{dL}{dW_o}$$

$$\text{softmax}(Q \cdot K^T) \text{ for } \frac{dL}{dV}$$

(where we have ignored the layernorm operations)

Computational graph:

~~mm~~ = Saved.



Therefore we need :

$$\text{FLOPs} = 4 \text{BTSNH} + O(\text{BTF})$$

to rematerialize during the backward pass.

Question 7:

$$\text{FLOPs for V3} = 6 \cdot 37e9 \cdot 14.8e12$$

$$\text{H800 SXM FP8 Tensor Core FLOPs/s} = 1474 \text{ TFLOPs/s}$$

Available FLOPs in 2.74M H800 hours

$$\text{if using FP8} = 2.79e6 \cdot 3600 \cdot 1474e12$$

$$\text{Hardware Utilization} = \frac{6 \cdot 37 \cdot 14.8 \cdot 10^{21}}{2.79 \cdot 3600 \cdot 1474 \cdot 10^{18}}$$

$$= \frac{6 \cdot 37 \cdot 14.8}{2.79 \cdot 3600 \cdot 1474} \cdot 10^3 \approx 0.00016 \cdot 10^3 = 0.16.$$

Question 8:

Assume training flops. E copies of standard MLP blocks, and each token activates K of these experts. Consider the following model of MOE:

$$h_e = \sum_{i=1}^E g_{i,e} \text{FFN}_i(x_e)$$

where $\text{FFN}_i(x_e) = W_i \cdot x_e$ and $W_i \in \mathbb{R}^{d \times d}$.

To compute the MOE layer we will group the tokens belonging to an expert and then matmul. So in total we will do E matmuls. Assume that every token randomly chooses K numbers from $\{1, \dots, E\}$.

Define :

$$S_e = \sum_{i=1}^T X_{i,e}$$

where $X_{i,e} = 1$ [Token i has sampled $e \in \{1 \dots E\}$].

The expected value of S_e gives us the expected value of the batch dimension of the matmul of expert e :

$$E[S_e] = \sum_{i=1}^T P(X_{i,e}) = \sum_{i=1}^T \frac{k}{E} \approx \frac{T \cdot k}{E}$$

So then we do:

For e in $\{1 \dots E\}$:

$$1: \left[\frac{T \cdot k}{E}, D \right] @ \text{int8} [D, D]$$

$$\text{FLOPs} = E \cdot \left(2 \cdot \frac{T \cdot k}{E} \cdot D^2 \right) \approx 2T \cdot k \cdot D^2$$

$$\text{Bytes} = E \cdot \left(\frac{2 \cdot T \cdot k \cdot D}{E} + D^2 \right) = 2TkD + D^2$$

$$AI = \frac{2 \cdot T \cdot k \cdot D^2}{2 \cdot T \cdot k \cdot D + E D^2} \approx \frac{2 \cdot T \cdot k \cdot \cancel{D^2}}{E \cdot \cancel{D^2}}$$

$$= \frac{2 \cdot T \cdot k}{E}$$

Therefore for being compute bound:

$$\frac{2 \cdot T \cdot k}{E} \geq 480$$

$$\Rightarrow T \geq \frac{E}{k} \cdot \frac{480}{2} = \frac{E}{k} \cdot 240$$

So for DeepSeek V3 this number is 7680.